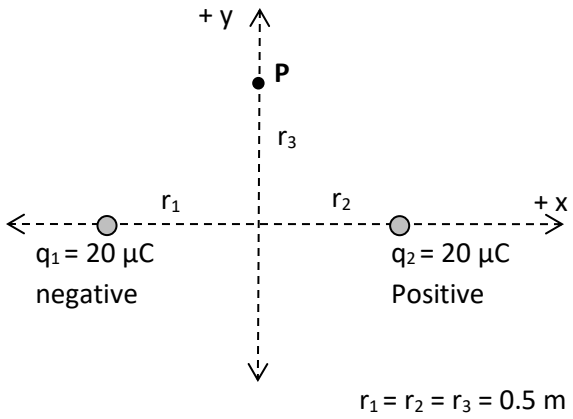
**Worksheet # 2 Part B - Upload Solutions**
**Physics 1**

Name: _____ Date: _____

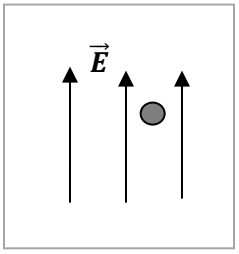
**Must show equations used and steps taken in determining the solution for each problem! No work No credit.**

**Upload the solutions to Questions 11 though 15**

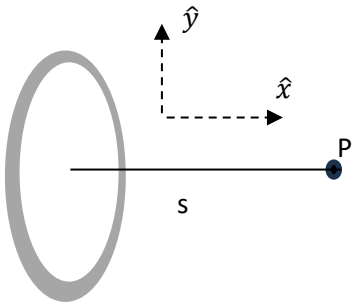
11) Using the diagram below, determine the magnitude and direction (α and β) of the net electric field at point P. The net charge is the same for both spheres q = 20 μC but $q_1$ is a negative charge and $q_2$ is a positive charge. The distance from the origin to the charge and point P is the same and is $r_1 = r_2 = r_3 = 0.5$ m




12) A ping pong ball (m = 0.005 kg) has a net charge of q, it is placed in a vertical electric field with a strength of 1,000 N/C directed straight up. What net charge q and type of charge (positive or negative) must the ping ball have in order for it to stay stationary in the gravitational field?

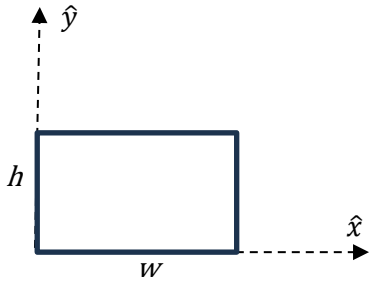


13) A circular thin wire with a radius R and a uniform linear charge density of $\lambda$ is shown below, determine the electric field along the x-axis at point P which is at a distance of s from the center of the wire. Draw and label the diagram, the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.



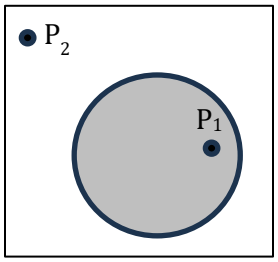

14) The electric field magnitude changes along the y-axis by the equation below, determine the equation for the electric flux through the rectangle shown below.

$$\vec{E}(x) = \left(my^{1/2}\right)\hat{z}$$



15) A non-conducting sphere of radius R = 30 cm has a volume charge density of $\rho = 4 \times 10^{-6}$ C/m$^3$ uniformly distributed throughout its volume.

A) Determine the electric field strength at point $P_1$, that is at a distance of $r_1 = 20$ cm from the center of the sphere.




B) Determine the electric field strength at point $P_2$, that is at a distance of $r_2 = 100$ cm from the center of the sphere.